Ch 12: Magnetic Effects of Electric Current

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SECTION A - INTRODUCTION TO MAGNETISM, ELECTRIC CURRENT AND MAGNETIC FIELD

1. Magnetic Effect of Electric Current

- When electric current flows through a conductor, it produces a **magnetic field** around it.
- This phenomenon is called the **magnetic effect of electric current**.
- It shows that electricity and magnetism are closely related.

2. Magnets and Their Properties

- A **magnet** is a material that attracts magnetic materials such as iron, nickel, cobalt, and steel.
- A magnet has two poles: North Pole (N) and South Pole (S).
- Like poles repel; unlike poles attract.
- Magnetic poles always occur in pairs. Cutting a magnet gives smaller magnets, each with both N and S poles.

Pole Behaviour

- Like poles (N–N or S–S) → Repel each other.
- Unlike poles (N–S) → Attract each other.

3. Magnetic Field

- The region around a magnet or a current-carrying conductor where a magnetic force can be experienced is called a **magnetic field**.
- It is a vector field; it has both magnitude and direction.
- The strength of the magnetic field decreases as we move away from the source.

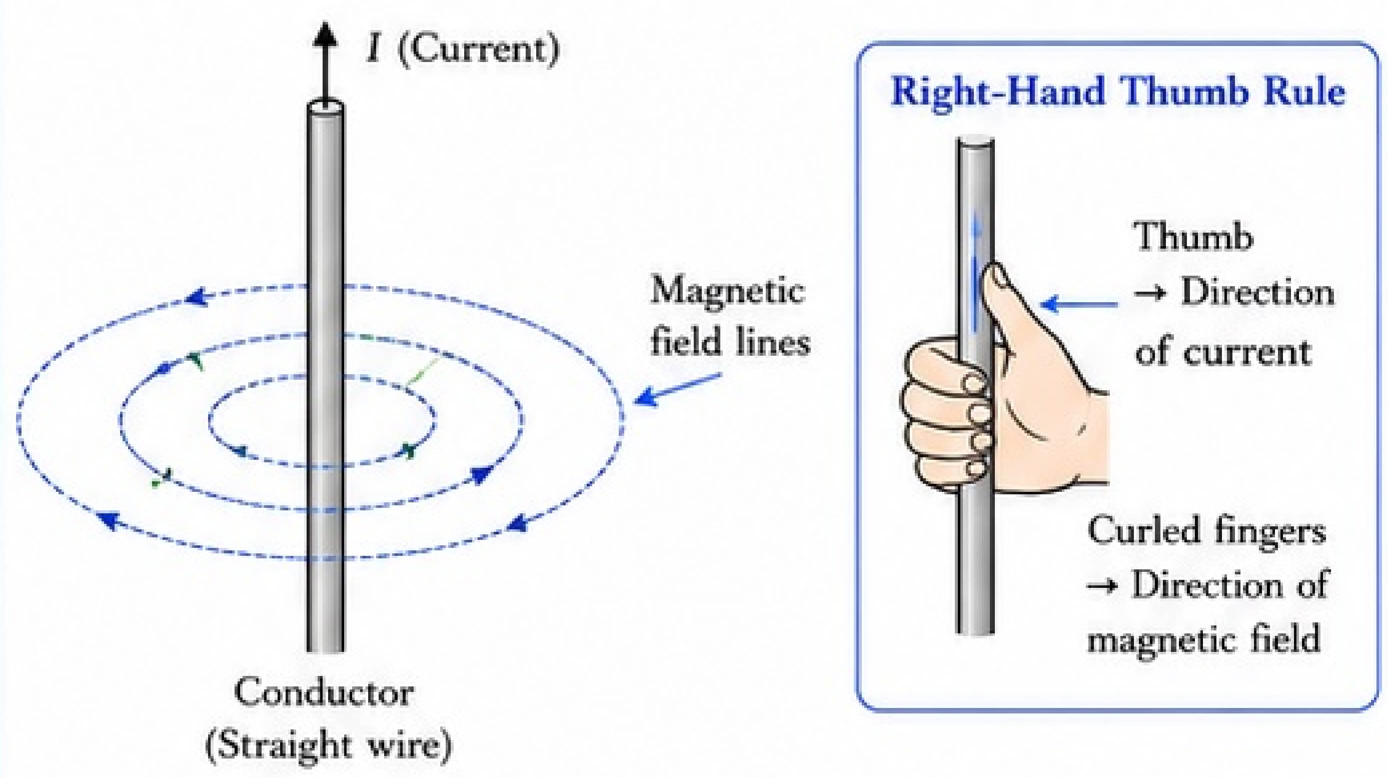
4. Magnetic Field Lines and Their Properties

- Magnetic field is represented by imaginary lines called **magnetic field lines.**
- Properties of magnetic field lines:
  - (i) They are continuous closed curves.
  - (ii) Outside a magnet, they go from North pole to South pole.
  - (iii) Inside a magnet, they go from South pole to North pole.
  - (iv) The tangent at any point on a field line gives the direction of the magnetic field at that point.
  - (v) No two field lines intersect each other.
  - (vi) The closer the lines, the stronger is the field.

5. Magnetic Field Around a Straight Current-Carrying Conductor

- When current flows through a straight conductor, concentric circular magnetic field lines are produced around it.
- The direction of the field lines is given by the **Right-Hand Thumb Rule:**
- Hold the conductor in your right hand with the thumb pointing in the direction of current. The curled fingers show the direction of magnetic field.
- The stronger the current, the stronger the magnetic field.

Magnetic Field Around a Straight Current-Carrying Conductor



Important Note: The magnetic field lines are concentric circles centered on the conductor. The direction depends on the direction of current.

Magnet vs Magnetic Field

Aspect	Magnet	Magnetic Field
Nature	Material object	Region around magnet or current
Has Poles?	Yes (N and S poles)	No poles
Visible?	Yes (magnet is visible)	No (invisible)
Representation	Bar magnet, horseshoe magnet, etc.	Field lines
Effect	Attracts or repels magnetic materials	Exerts force on a moving charge or magnetic pole
Depends on	Material and shape	Source (magnet or current) and distance

Important Terms to Remember

- **Magnet:** A material that produces a magnetic field and attracts magnetic materials.
- **Magnetic Field:** The region around a magnet or a current-carrying conductor where a magnetic force acts.
- **Magnetic Field Lines:** Imaginary lines used to represent magnetic field; they show the direction and strength of the field.
- **Current-Carrying Conductor:** A conductor through which electric current is flowing.

QUICK RECALL

- Current in a conductor → magnetic field around it.
- Magnet has two poles: N and S.
- Like poles repel; unlike poles attract.
- Field lines are closed loops: outside N → S, inside S → N.
- Around a straight wire, field lines are concentric circles.

KEY IDEA

Electric current and magnetism are connected. A current-carrying conductor behaves like a magnet, creating a magnetic field around it. Understanding magnetic field lines and their direction helps us predict magnetic effects in different situations.

EXAM TIPS

- Define terms clearly: magnet, magnetic field, field lines.
- For diagrams, show concentric circles with arrow direction correctly.
- State Right-Hand Thumb Rule with diagram.
- Write properties of field lines in points.

CENTUM FOCUS

- Memorize properties of field lines.
- Learn Right-Hand Thumb Rule with neat diagram.
- Practice drawing concentric field lines for a straight conductor.
- Understand and revise all definitions and key terms.

SECTION B - MAGNETIC FIELD DUE TO CURRENT IN A CIRCULAR LOOP AND A SOLENOID

1. Magnetic Field Due to a Circular Current-Carrying Loop

- When current flows in a circular loop, a magnetic field is produced around it.
- The field lines are concentric near the loop and resemble a bar magnet.
- At the centre of the loop, the field is strong and perpendicular to the plane of the loop.
- The magnitude of the field at the centre of a single loop is:

$$B = \frac{\mu_0 I}{2R}$$

where, μ_0 = Permeability of free space
I = Current in the loop
R = Radius of the loop

2. Direction of Magnetic Field of a Circular Loop

- The direction of the magnetic field at the centre of the loop is given by the **Right-Hand Thumb Rule**.
- Curl the fingers of your right hand in the direction of current in the loop.
- The thumb then points in the direction of the magnetic field at the centre (normal to the plane of the loop).

3. Effect of Number of Turns and Current

- For a coil of N turns (circular coil), the magnetic field at the centre is:

$$B = \frac{\mu_0 N I}{2R}$$

where, N = Number of turns

- More number of turns (N) $\longrightarrow$ stronger magnetic field.
- More current (I) $\longrightarrow$ stronger magnetic field.
- Smaller radius (R) $\longrightarrow$ stronger magnetic field.

4. Magnetic Field Due to a Solenoid

- A solenoid is a long coil of many circular turns of insulated copper wire wound closely in the shape of a cylinder.
- When current flows through a solenoid, it produces a magnetic field similar to that of a bar magnet.
- Field inside the solenoid is strong and uniform.
- Field outside the solenoid is weak.
- Magnitude of magnetic field inside a long solenoid:

$$B = \mu_0 n I$$

where, n = Number of turns per unit length

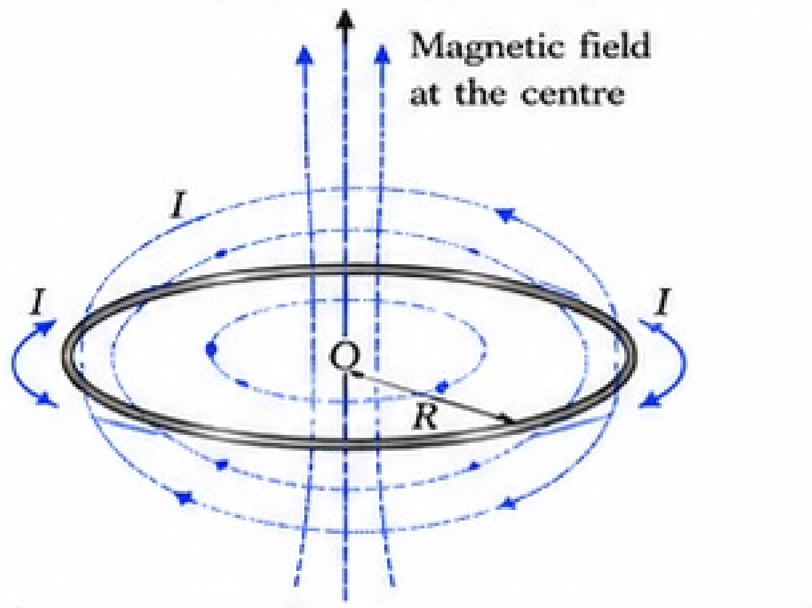
5. Why Does a Solenoid Behave Like a Bar Magnet?

- Field lines inside a solenoid are parallel, close, and strong like inside a bar magnet.
- Field lines outside the solenoid emerge from one end and enter the other end.
- Hence, one end acts as North Pole and the other as South Pole.
- The pole at which field lines emerge is called North Pole.
- The pole at which field lines enter is called South Pole.

6. Key Points to Remember

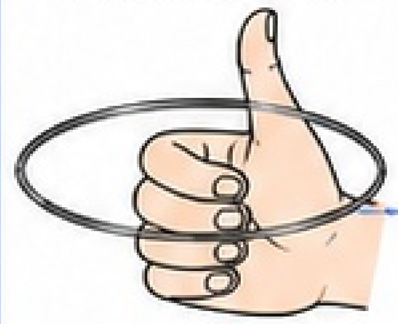
- Magnetic field is a vector quantity.
- The strength and direction of magnetic field depend on current, number of turns, and geometry of the conductor.
- Solenoid is widely used in electromagnets.

Magnetic Field Due to a Circular Current-Carrying Loop



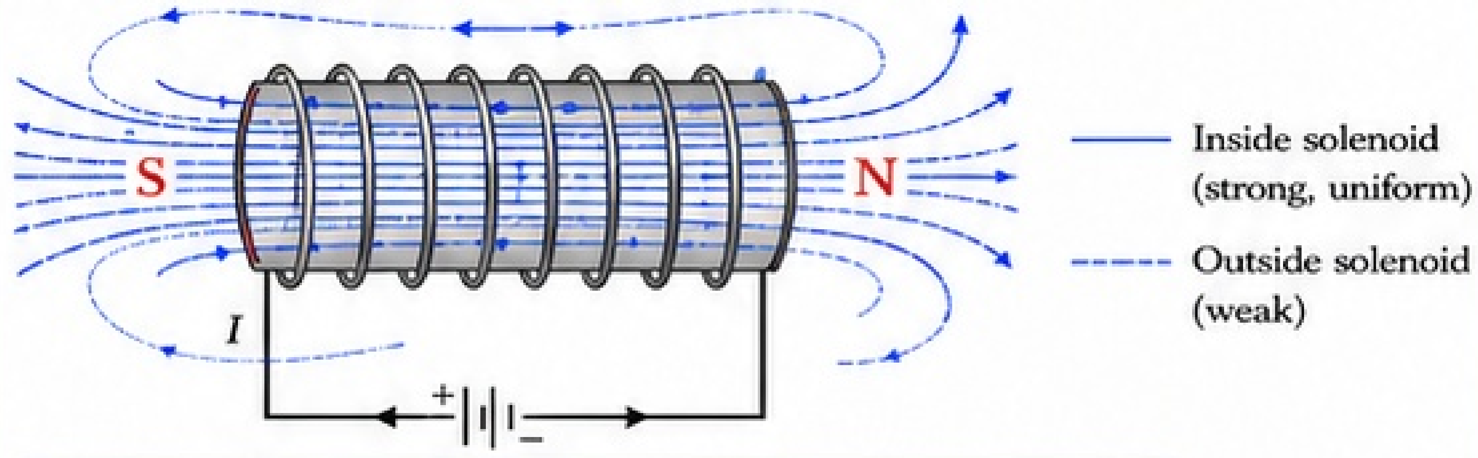
Right-Hand Thumb Rule (for Loop)

Thumb → B (out of the plane)



Important Note: If the current appears anticlockwise, the magnetic field at the centre is out of the plane of the loop. If current appears clockwise, the field is into the plane.

Magnetic Field Due to a Solenoid



Note: The end where current appears anticlockwise is North Pole. The end where current appears clockwise is South Pole.

Comparison: Straight Conductor, Circular Loop and Solenoid

Aspect	Straight Conductor	Circular Loop (Single Turn)	Solenoid (Long Coil)
Shape	Long straight wire	Circular coil (one turn)	Cylindrical coil (many turns)
Magnetic Field Pattern	Concentric circles around the wire	Concentric circles near the loop; bar magnet-like pattern	Like a bar magnet inside and outside
Field Direction at Centre	Not applicable	Perpendicular to the plane of loop	From South to North inside the solenoid
Field Strength	Weak	Stronger (depends on I, R, N)	Strong and uniform inside
Ends	No poles	No distinct poles	North Pole and South Pole
Formula (Field at Centre/Inside)	$B = \frac{\mu_0 I}{2\pi r}$ (at distance r)	$B = \frac{\mu_0 N I}{2R}$	$B = \mu_0 n I$

Important Terms and Definitions

- **Circular Loop:** A coil of wire in a circular shape carrying current.
- **Solenoid:** A long coil of many turns of insulated wire wound in a cylindrical shape.
- **Magnetic Pole:** A region where the magnetic field is strongest.
- **North Pole:** End of a solenoid where field lines emerge.
- **South Pole:** End of a solenoid where field lines enter.
- **Field Line:** An imaginary line that represents the direction of magnetic field.

QUICK RECALL

- Circular loop: $B = \frac{\mu_0 N I}{2R}$
- Solenoid: $B = \mu_0 n I$
- More N / I ⇒ stronger field.
- Right-Hand Thumb Rule → direction of field.
- Solenoid behaves like a bar magnet.

KEY IDEA

- Magnetic field depends on shape of conductor and current.
- Loops produce field at the centre; solenoids produce uniform strong field inside.
- Solenoids are used in electromagnets.

EXAM TIPS

- Always draw field lines with correct direction.
- Remember formulas with units: B in tesla (T).
- State Right-Hand Thumb Rule clearly in answers.
- Link number of turns and current with field strength.

CENTUM FOCUS

- Practice diagrams of loop and solenoid with field lines.
- Learn derivations and formulas perfectly.
- Solve numericals using $B = \frac{\mu_0 N I}{2R}$ and $B = \mu_0 n I$.
- Revise differences and definitions regularly.

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SECTION C - FORCE ON A CURRENT-CARRYING CONDUCTOR IN A MAGNETIC FIELD

1. Effect of Magnetic Field on a Current-Carrying Conductor

- When a current-carrying conductor is placed in a magnetic field, it experiences a force.
- This phenomenon is called the **motor effect**.
- The conductor moves in the direction of the force.

2. Conditions for Force (Motor Effect)

- A magnetic field must be present.
- Electric current must flow through the conductor.
- The conductor must be placed such that it is not parallel to the magnetic field.
- If current is parallel or anti-parallel to the magnetic field, no force acts on the conductor.

3. Direction of Force

- The direction of the force depends on:
  (i) Direction of magnetic field (N → S)
  (ii) Direction of current
  (iii) Orientation of the conductor
- These three are mutually perpendicular.

4. Fleming's Left-Hand Rule

- Used to find the direction of force on a current-carrying conductor in a magnetic field.
- Stretch the thumb, forefinger and middle finger of your left hand such that they are mutually perpendicular.
- Forefinger → Direction of Magnetic Field (N → S)
- Middle Finger → Direction of Current
- Thumb → Direction of Force (Motion)

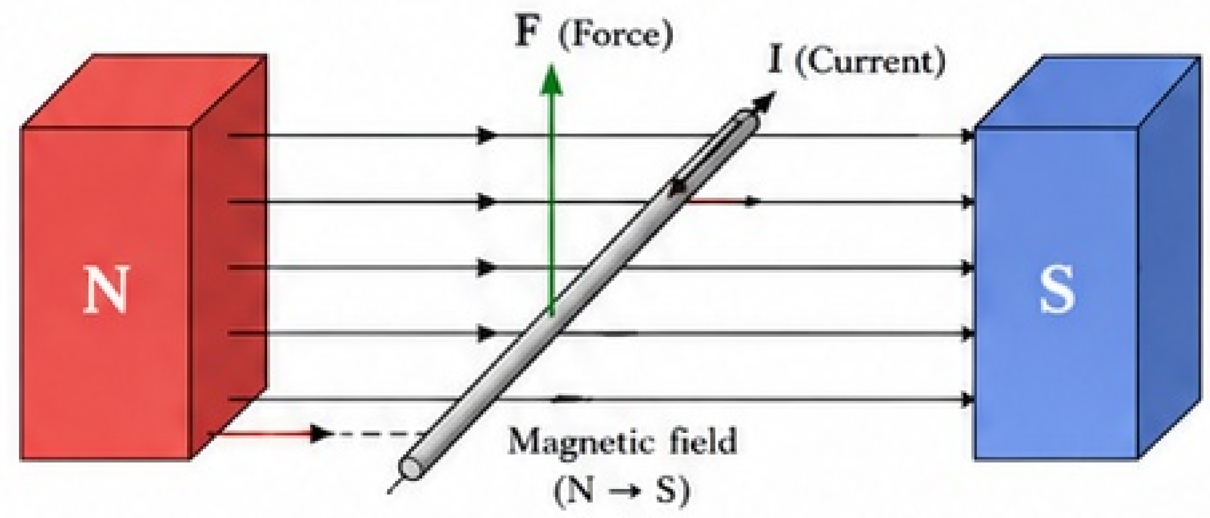
5. Factors Affecting the Force on a Conductor

- The magnitude of force depends on:
  (i) Strength of magnetic field (B)
  (ii) Strength of current in the conductor (I)
  (iii) Length of the conductor in the field (l)
  (iv) Angle (θ) between the direction of current and field
- Maximum force acts when conductor is perpendicular ($\theta = 90°$) to the magnetic field.

Formula: $F = BIl \sin\theta$

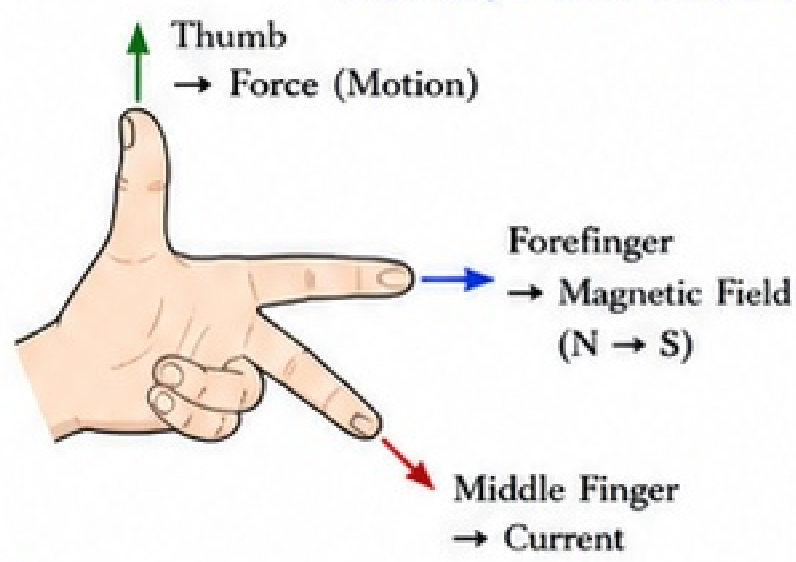
where,
F = Force (N), B = Magnetic field (tesla, T)
I = Current (A), l = Length in field (m), θ = Angle

Current-Carrying Conductor in a Magnetic Field



If current flows from left to right and magnetic field is from N to S, the conductor experiences an upward force.

Fleming's Left-Hand Rule



Remember:
- Forefinger → Field
- Middle Finger → Current
- Thumb → Force

Effect of Direction of Field and Current on Force

S. No.	Magnetic Field	Current	Result / Effect (Force Concept)
1.	N → S (Left to Right)	Left to Right	Force acts Upwards
2.	N → S (Left to Right)	Right to Left	Force acts Downwards
3.	S → N (Right to Left)	Left to Right	Force acts Downwards
4.	S → N (Right to Left)	Right to Left	Force acts Upwards

Practical Significance (Motor Effect)

- A current-carrying conductor experiences force in a magnetic field.
- This principle is used in **electric motors, loudspeakers, moving-coil instruments**, and many electrical devices.
- It converts electrical energy into mechanical energy.

Concept Summary

- A current-carrying conductor in a magnetic field experiences a force.
- Direction of force is given by Fleming's Left-Hand Rule.
- Force is maximum when conductor is perpendicular to the field.
- Basis of working of electric motors and many practical devices.

QUICK RECALL

- **Motor effect**: Force on conductor carrying current in magnetic field.
- **Conditions**: Magnetic field, current and non-parallel placement.
- **Left-Hand Rule**: Forefinger → Field, Middle Finger → Current, Thumb → Force.
- **Force**: $F = BIl \sin\theta$

KEY IDEA

Force acts only when there is interaction between magnetic field and current. Changing the direction of either field or current reverses the direction of force.

EXAM TIPS

- Always draw a neat diagram with N and S poles.
- Identify directions clearly before applying the rule.
- Remember: Left-Hand Rule (Not Right-Hand).
- Use $F = BIl \sin\theta$ in numericals.

CENTUM FOCUS

- Practice Left-Hand Rule with various diagrams.
- Learn direction table by heart.
- Solve numericals using $F = BIl \sin\theta$.
- Relate motor effect to examples in daily life.

SECTION D - ELECTRIC MOTOR: PRINCIPLE, CONSTRUCTION AND WORKING

1. What is an Electric Motor?

- An electric motor is a device that converts electrical energy into mechanical energy. It produces rotation when current flows through a coil placed in a magnetic field.

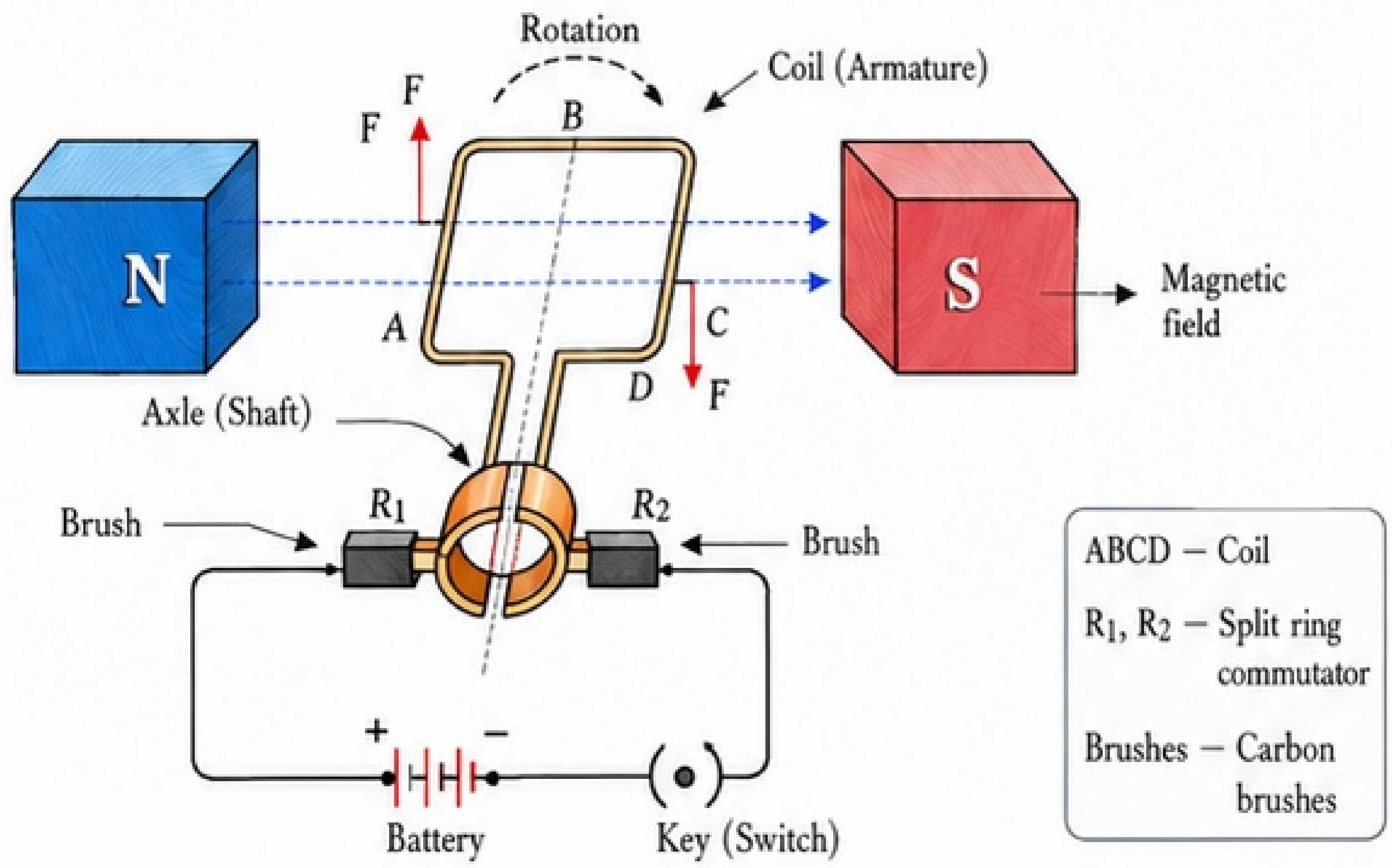
2. Principle of Electric Motor

- A current-carrying conductor placed in a magnetic field experiences a force.
- A rectangular coil carrying current in a magnetic field experiences forces on its two opposite sides in opposite directions. These forces form a couple and produce a torque which rotates the coil.

3. Construction of a Simple DC Motor

- The main parts are shown in the diagram.
- All parts work together to produce continuous rotation.

Construction of a Simple DC Motor



Components of a Simple DC Motor and Their Functions

Component	Function
Battery	Provides electrical energy to the circuit.
Key (Switch)	Completes or breaks the circuit.
Brushes (Carbon brushes)	Provide contact between the rotating commutator and the external circuit.
Split Ring Commutator	Reverses the current in the coil after every half turn to keep rotation in the same direction.
Armature / Coil	Rectangular coil where current flows; it experiences torque and rotates.
Axle (Shaft)	Supports the coil and allows it to rotate freely.
Magnet Poles (N and S)	Provide a strong magnetic field in which the coil rotates.
Frame / Stand	Supports all the parts of the motor.

4. Working of a Simple DC Motor – Step by Step

1. When the key is closed, current flows from the battery through the brushes, split ring commutator and the coil (armature).
2. In the magnetic field (from N to S), the two sides of the coil experience forces in opposite directions.
3. These opposite forces form a couple which turns the coil.
4. After half a turn, the split ring commutator reverses the direction of current in the coil.
5. Due to reversal, the directions of forces on the coil sides also reverse.
6. The couple continues to act in the same rotational direction.
7. This process repeats continuously, and the coil keeps rotating.

5. Role of Split Ring Commutator

- It automatically reverses the direction of current in the coil after every half rotation.
- This ensures that the couple always acts in the same rotational direction, so the coil keeps rotating.

6. Why does the Direction of Current Reverse?

- After half a turn, the coil sides interchange their positions.
- If current did not reverse, the forces would also reverse and the coil would start rotating in the opposite direction.
- The split ring reverses the current, keeping the forces in the same rotational sense.

7. How is Energy Converted?

- Electrical energy from the battery → current in the coil.
- Magnetic field + current → force on coil sides.
- Force produces torque → rotates the coil.
- Hence, electrical energy is converted into mechanical (rotational) energy.

8. Uses of Electric Motors in Daily Life

- **Electric fans** – rotate the blades and provide airflow.
- **Mixers and grinders** – rotate the blades to mix and grind.
- **Washing machines** – rotate the drum for washing.
- **Water pumps** – draw water by rotating the pump.
- **Electric toys and cars** – provide motion.
- **Power tools** – drills, saws, screwdrivers, etc.
- **Many industrial machines** – lathes, conveyor belts, cranes, etc.

QUICK RECALL

- Motor works on the force on a current-carrying conductor.
- Coil in magnetic field → couple → rotation.
- Split ring reverses current after every half turn.
- Electrical energy → mechanical energy.

KEY IDEA

The split ring commutator is the heart of a DC motor. It reverses the current at the right moment so that the torque always acts in the same direction, ensuring continuous rotation.

EXAM TIPS

- Always draw a neat labelled diagram of a simple DC motor.
- Explain construction, working and role of split ring clearly.
- Mention the energy conversion.
- 5–7 steps in working sequence fetch full marks.

CENTUM FOCUS

- Memorize components and their functions.
- Understand why current must reverse.
- Relate motors with real-life applications.
- Practice diagram labelling for accuracy.

SECTION E - ELECTROMAGNETIC INDUCTION AND FLEMING'S RIGHT-HAND RULE

1. Electromagnetic Induction – Meaning

- When there is a change in magnetic field linked with a conductor or coil, an electric current is induced in it.
- This phenomenon is called **electromagnetic induction.**

2. Faraday's Discovery

- In 1831, Michael Faraday discovered that changing magnetic conditions produce electric current.
- Faraday's experiments laid the foundation of transformer, induction motor, generator, etc.

3. How Does Changing Magnetic Field Induce Current?

- A change in magnetic field produces an induced emf in a conductor.
- If the circuit is closed, this emf produces an induced current.
- Greater the rate of change of magnetic field, stronger is the induced current.

4. Induction by Moving a Magnet Near a Coil

- When a bar magnet is moved towards a coil connected to a galvanometer, the galvanometer shows deflection (current induced).
- When the magnet is moved away, deflection occurs in the opposite direction.
- If the magnet is kept stationary, no current is induced.
- Faster motion → greater deflection.

5. Induction by Changing Current in a Nearby Coil

- When current in a coil changes (switched ON or OFF), the magnetic field around it also changes.
- This changing magnetic field induces current in a nearby coil.
- No current is induced if the current in the primary coil is steady and constant.

6. Factors Affecting the Magnitude of Induced Current

- The magnitude of induced current depends on the following:

Factors that Increase or Decrease Induced Current

Factor	Increase Induced Current	Decrease Induced Current
Rate of change of magnetic field	Change rapidly	Change slowly
Number of turns in coil	More turns	Fewer turns
Area of the coil	Larger area	Smaller area
Strength of magnetic field	Stronger magnet	Weaker magnet
Relative motion	Move faster	Move slower
Orientation (angle)	Coil perpendicular to field	Coil parallel to field

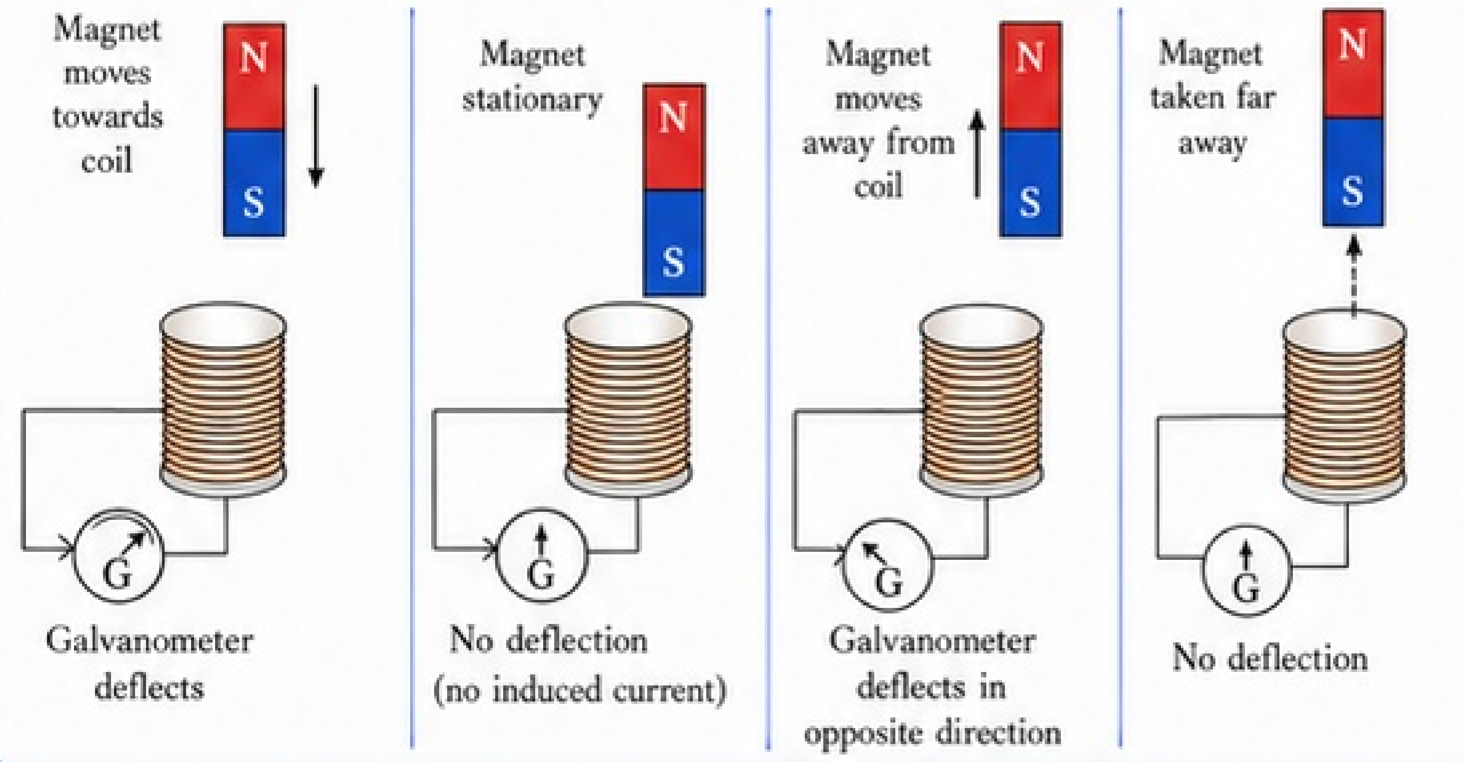
7. Direction of Induced Current

- The direction of induced current is given by Fleming's Right-Hand Rule.
- It always acts in such a way that it opposes the change that produced it (Lenz's Law).

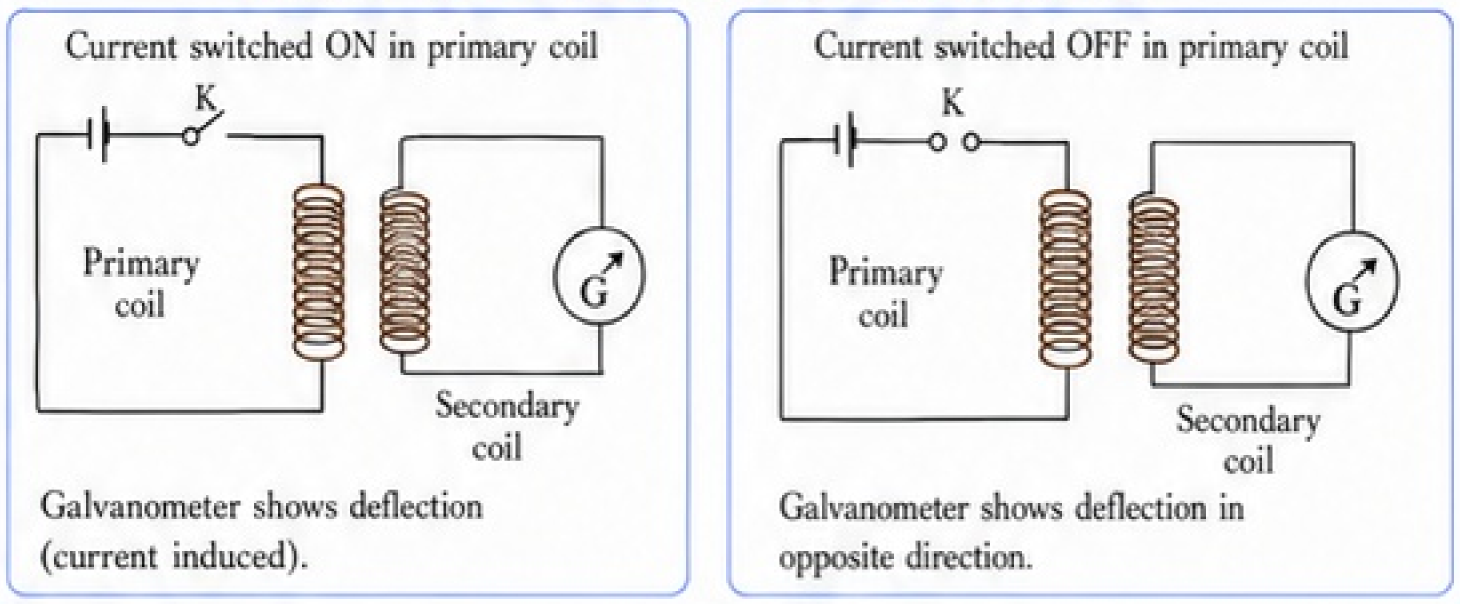
8. Fleming's Right-Hand Rule

- Hold the right hand with thumb, forefinger and middle finger mutually perpendicular.
- Forefinger → direction of magnetic field.
- Thumb → direction of motion of conductor.
- Middle finger → direction of induced current.

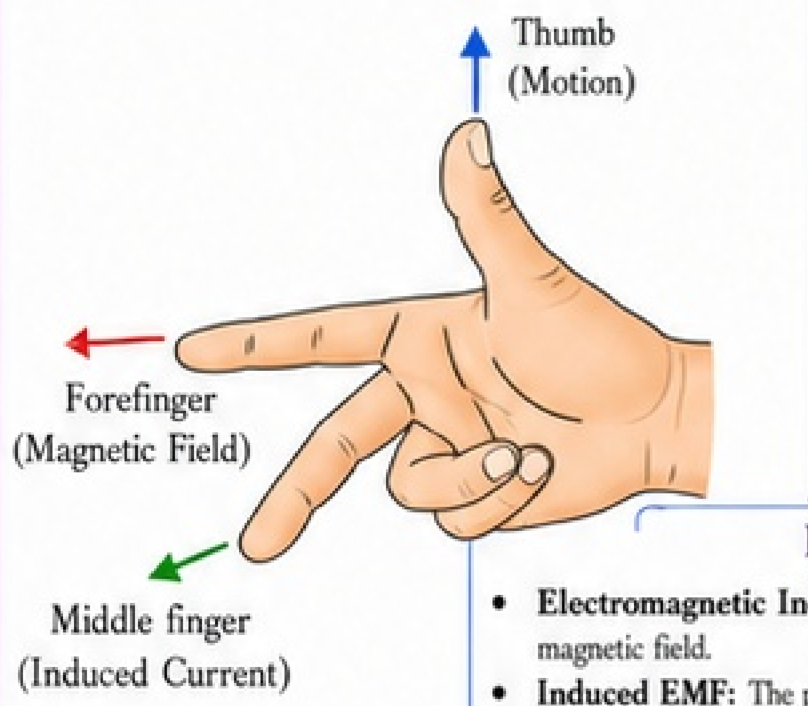
Induction by Moving a Magnet Into and Out of a Coil



Induction by Changing Current in a Nearby Coil



Fleming's Right-Hand Rule



Practical Meaning

- Electromagnetic induction is the principle behind generators, transformers, induction motors, wireless charging, etc.
- It is used to convert mechanical energy into electrical energy.
- It enables the production and transmission of electricity.

Important Terms to Remember

- **Electromagnetic Induction:** Production of current due to change in magnetic field.
- **Induced EMF:** The potential difference produced in a conductor due to change in magnetic field.
- **Induced Current:** The current produced when the circuit is closed.
- **Lenz's Law:** The induced current opposes the change that produces it.
- **Fleming's Right-Hand Rule:** Rule to find the direction of induced current.

QUICK RECALL

- Change in magnetic field → induced emf → induced current.
- Move magnet → current induced.
- Change current → current induced.
- Greater change → greater current.
- Fleming's Right-Hand Rule gives direction.

KEY IDEA

A changing magnetic field is the key to inducing current. The more rapid the change and the better the setup (more turns, larger area, stronger field), the greater will be the induced current.

EXAM TIPS

- Draw clear diagrams showing movement and deflection.
- State Fleming's Right-Hand Rule with neat hand diagram.
- List factors increasing/decreasing induced current.
- Use Lenz's Law to find direction when asked.

CENTUM FOCUS

- Practice numericals on factors affecting induced current.
- Compare induction by moving magnet and changing current.
- Relate induction to practical appliances.
- Revise rules and definitions thoroughly.

SECTION F - ELECTRIC GENERATOR: PRINCIPLE, CONSTRUCTION, WORKING AND AC/DC

1. Principle of Electric Generator

- A generator works on the principle of **electromagnetic induction**.
- When the armature coil rotates in a magnetic field, the magnetic flux linked with the coil changes.
- Due to this change in flux, an emf is induced in the coil.
- If the circuit is complete, induced current flows in the coil.

Relation to Electromagnetic Induction

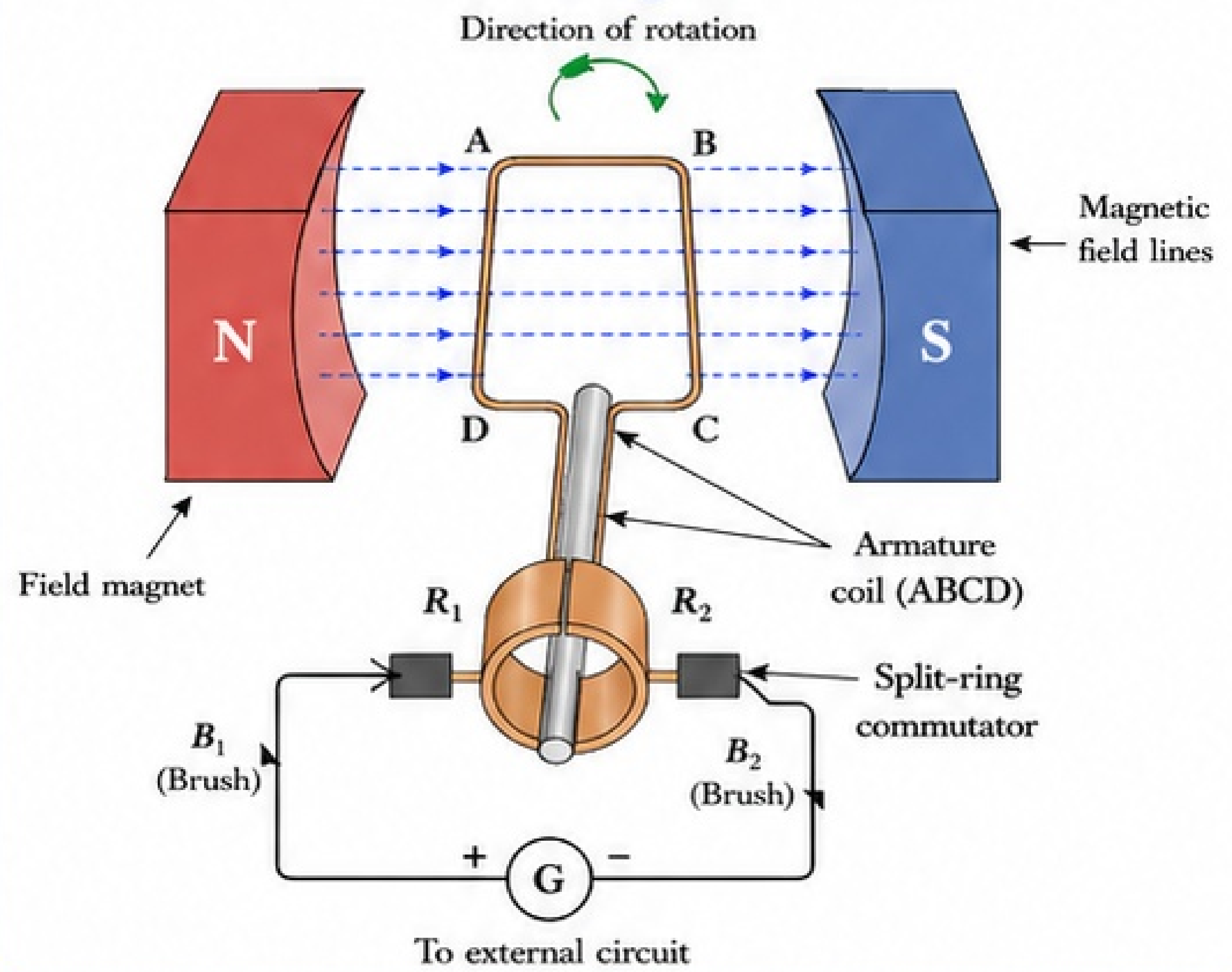
- Based on Faraday's Law of Electromagnetic Induction:

$$E = -N\frac{d\Phi}{dt}$$

Where,
E = Induced emf (volt)
N = Number of turns
$\frac{d\Phi}{dt}$ = Rate of change of magnetic flux (weber/s)

- The direction of induced emf is given by Fleming's Right-Hand Rule.

Construction of a Simple DC Generator



2. Construction of a Simple DC Generator

- It consists of the following main parts:
  - (i) **Field Magnet**: Produces magnetic field (N and S poles).
  - (ii) **Armature Coil**: Rectangular coil ABCD placed between poles.
  - (iii) **Shaft**: Connects the coil to the external mechanical source.
  - (iv) **Split-Ring Commutator**: Reverses the connection of coil after every half turn.
  - (v) **Carbon Brushes**: Two stationary brushes (B_1, B_2) in contact with the commutator to collect current.

Mechanical Energy to Electrical Energy Conversion

- A mechanical source (turbine, engine, windmill, etc.) rotates the coil.
- The rotational mechanical energy causes change in magnetic flux.
- This change in flux induces emf and produces electrical energy.
- Thus, the generator converts mechanical energy into electrical energy.

3. Working of a Simple DC Generator (Step by Step)

- **Step 1** → When the armature coil ABCD is rotated mechanically in the magnetic field from N pole to S pole, sides AB and CD cut magnetic field lines in opposite directions.
- **Step 2** → An emf is induced in the coil as per electromagnetic induction.
- **Step 3** → The induced current flows through the coil in one direction (say A → B → C → D).
- **Step 4** → After half rotation, the coil sides switch positions under opposite poles. The induced current in the coil reverses.
- **Step 5** → The split-ring commutator also reverses the connection of the coil with the external circuit.
- **Step 6** → Hence, the current in the external circuit flows in the same direction throughout the rotation.
- **Step 7** → Continuous rotation of the coil produces a unidirectional (DC) current.

4. AC Generator vs DC Generator

Aspect	AC Generator (Alternator)	DC Generator
1. Output	Alternating current (AC)	Direct current (DC)
2. Commutator	No commutator	Has split-ring commutator
3. Slip rings	Uses two slip rings	No slip rings
4. Brushes	Two carbon brushes connected to slip rings	Two carbon brushes in contact with commutator
5. Direction of current	Changes direction periodically	Unidirectional
6. Nature of emf	Alternating emf	Pulsating emf (but current is unidirectional)
7. Use	Used in power stations for large-scale power generation	Used in battery charging, electroplating, DC motors, etc.

5. Uses of Generators

- **AC Generators**: Used in thermal, hydro and nuclear power plants to generate electricity for homes, industries and transport.
- **DC Generators**: Used in battery charging, welding, electroplating, electrolysis and laboratory supplies.

QUICK RECALL

- Generator works on electromagnetic induction.
- Mechanical rotation → change in magnetic flux → induced emf.
- DC generator uses commutator; AC generator uses slip rings.
- Conversion: Mechanical energy → Electrical energy.

KEY IDEA

- Faraday's law is the heart of a generator.
- Commutator makes the current unidirectional in DC generator.
- Slip rings allow continuous alternating output in AC generator.
- Direction is decided by Fleming's Right-Hand Rule.

EXAM TIPS

- Label all parts of the generator diagram clearly.
- Write the working in proper sequence steps.
- Differentiate AC and DC generator in tabular form.
- State the principle (electromagnetic induction) in every answer.

CENTUM FOCUS

- Draw neat labelled diagrams of DC and AC generators.
- Learn the difference table perfectly.
- Practice numericals based on:

$$E = -N\frac{d\Phi}{dt}$$

- Relate energy conversion in words and flow.

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SECTION G - DOMESTIC ELECTRIC CIRCUITS, LIVE WIRE, NEUTRAL WIRE AND EARTHING

1. Domestic Electric Supply – Basics
- Electricity in our homes is supplied by an A.C. source (alternating current) with voltage about 230 V and frequency 50 Hz.
- Power enters the house through service wires from the distribution line.
- The supply connects to the main fuse and energy meter.
- From there, it is distributed to different circuits and appliances.

2. Live Wire (Phase Wire)
- The live wire brings current from the power station to our home.
- It is at high potential (~230 V) with respect to earth.
- Touching the live wire can cause a severe electric shock.
- It is identified by red or brown insulation.

3. Neutral Wire
- The neutral wire provides the return path for current to the source.
- It is connected to earth at the power station.
- Its potential is nearly 0 V (with respect to earth).
- Normally current does not flow in the neutral wire under safe conditions.
- It is identified by black or blue insulation.

4. Earth Wire (Earthing Wire)
- The earth wire is not meant for carrying current during normal use.
- It connects the metal body of appliances to the earth.
- If an insulation fault occurs, current flows through the earth wire, preventing electric shock and protecting the user.
- It is identified by green or green–yellow insulation.

5. Purpose of Earthing
- Provides a low resistance path for fault current.
- Keeps the metal body of appliances at zero potential.
- Ensures safety by causing the fuse/MCB to blow/trip during faults.

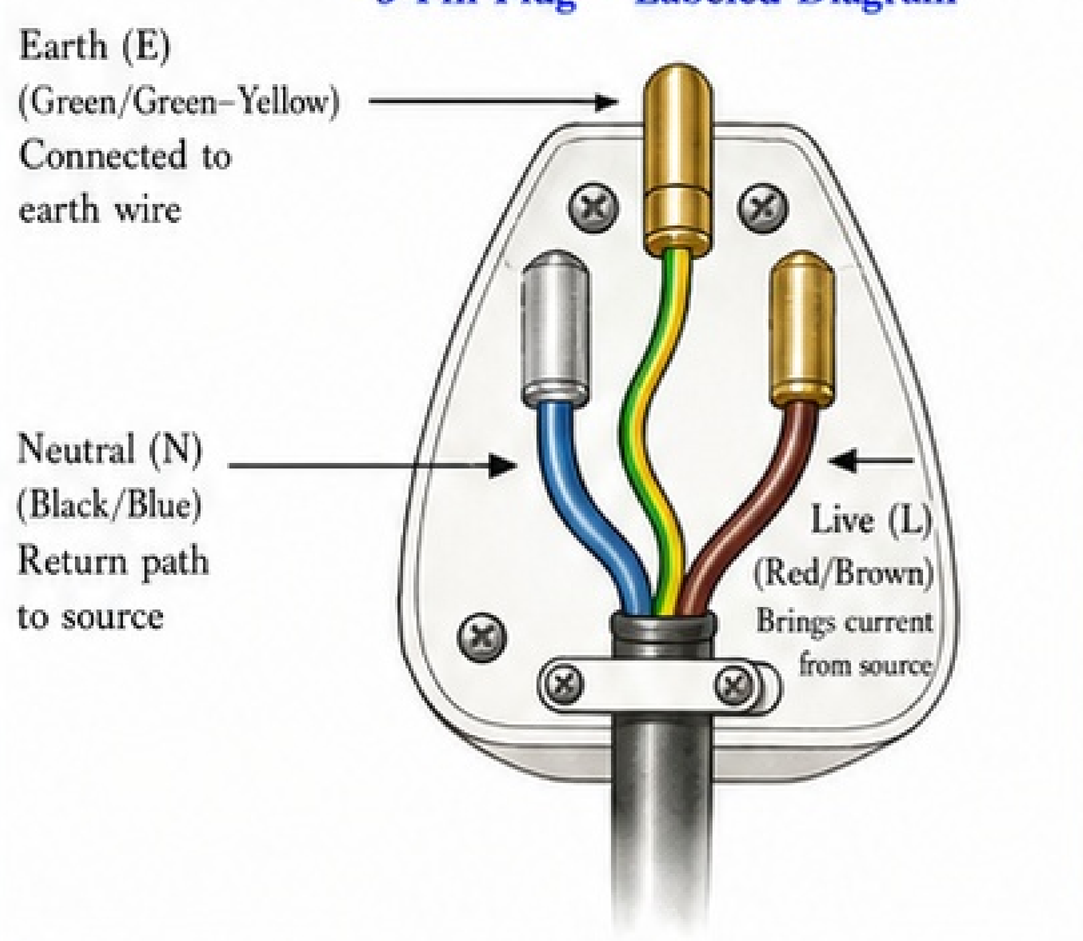
6. 3-Pin Plug
- A 3-pin plug has Live (L), Neutral (N) and Earth (E) pins.
- Live and neutral carry current; earth is for safety.
- Always connect the earth pin first and disconnect it last.

7. Why Earth Pin is Thicker and Longer?
- Earth pin is thicker so it makes contact before Live and Neutral.
- It is longer so it remains connected even when the plug is partially pulled out.
- This ensures safety by providing earthing before power connection.

8. Colour Coding of Wires (Basics)
- Standard colour code helps in safe installation and maintenance.
- Always follow the colour code for identification.

SAFETY FIRST
- Never touch live wire or a bare conductor.
- Do not use damaged wires or loose connections.
- Always use 3-pin plugs and ensure proper earthing.
- Switch off the main supply before repairing any appliance.
- In case of electric shock, switch off supply and seek help immediately.

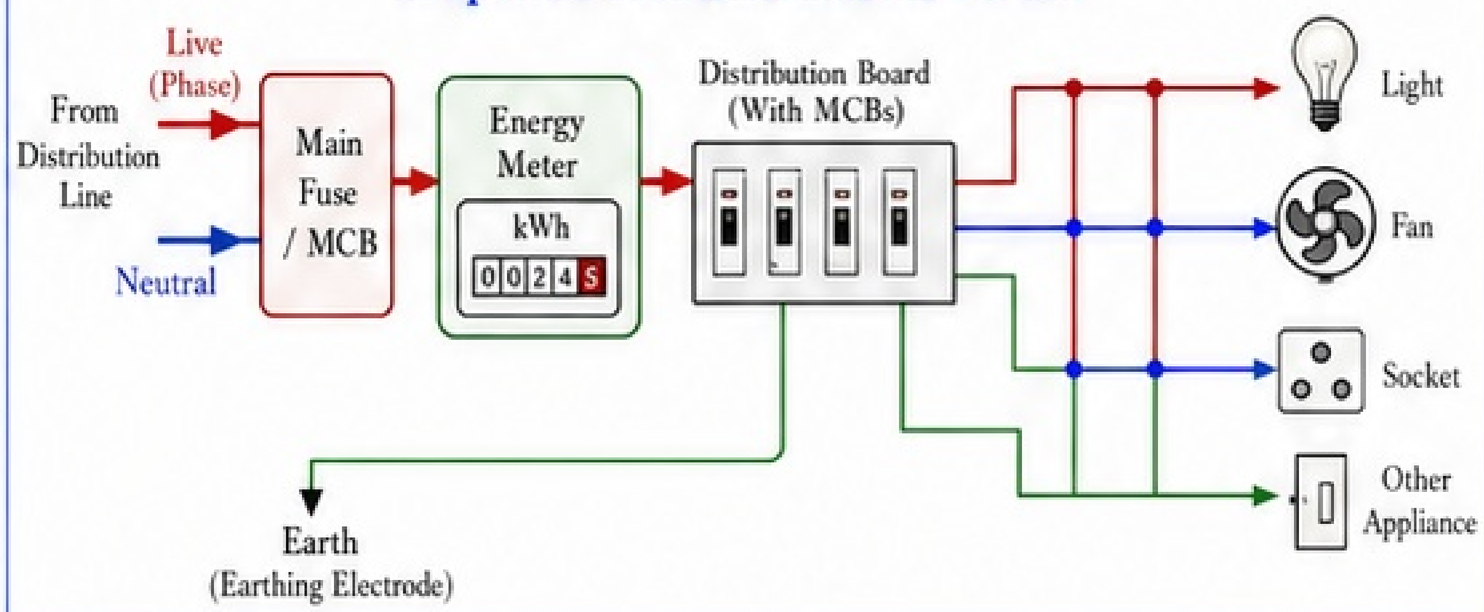
3-Pin Plug – Labeled Diagram



Key Points
- Earth pin is longer and thicker.
- It connects the appliance body to earth.
- Provides safety during faults.

Important Note: Never use a 2-pin plug for appliances with metal body. Always use a 3-pin plug and ensure proper earthing.

Simplified Domestic Electric Circuit



Wire Types and Functions

Wire	Colour Code (Common)	Function	Potential (w.r.t. Earth)
Live (Phase)	Red / Brown	Carries current from source to appliances	~230 V (High)
Neutral	Black / Blue	Returns current to source	~0 V (Low)
Earth (Earthing)	Green / Green–Yellow	Provides safety path for fault current	0 V (Earth)

QUICK RECALL
- Domestic supply: ~230 V, 50 Hz.
- Live brings current; Neutral returns it.
- Earth wire is for safety, not for normal current.
- 3-pin plug: L (Live), N (Neutral), E (Earth).
- Earth pin: longer & thicker for safety.

KEY IDEA

Live, Neutral and Earth together make the domestic electrical system safe and functional. Live carries power, Neutral completes the circuit, and Earth protects life and property. Always respect colour codes and use proper earthing.

EXAM TIPS
- Diagrams: Label L, N, E and show colour coding.
- Write functions of each wire clearly.
- Mention purpose of earthing in answers.
- Remember: Earth pin is longer and thicker.

CENTUM FOCUS
- Draw and label 3-pin plug with colour codes.
- Learn the domestic circuit diagram with main fuse, meter, MCB.
- Revise functions and safety rules regularly.
- Practise short answers: difference between live, neutral and earth.

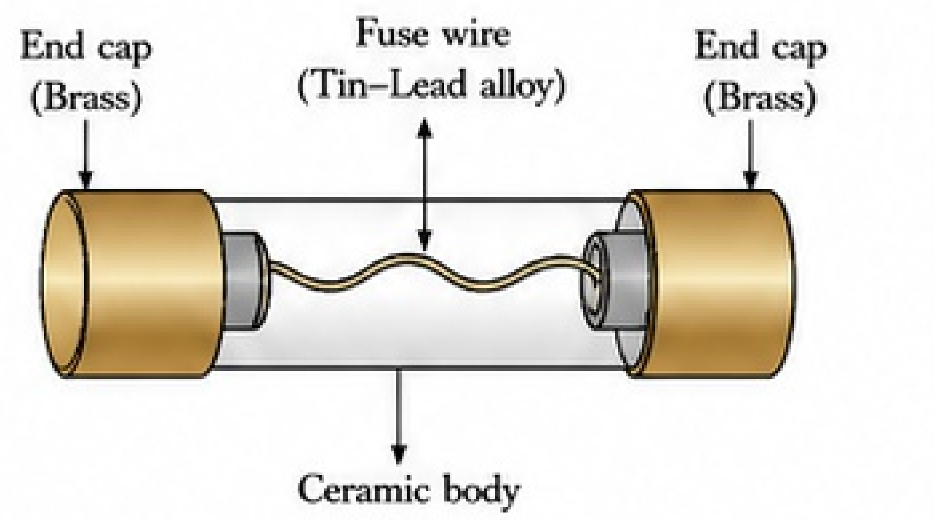
Ch 12: Magnetic Effects of Electric Current

Part 1 of 1 | Complete Notes | Deep Concepts | Diagrams | Tables | Exam Tips | Centum Focus

SECTION H – ELECTRIC FUSE, MCB, SHORT CIRCUIT, OVERLOADING AND SAFETY

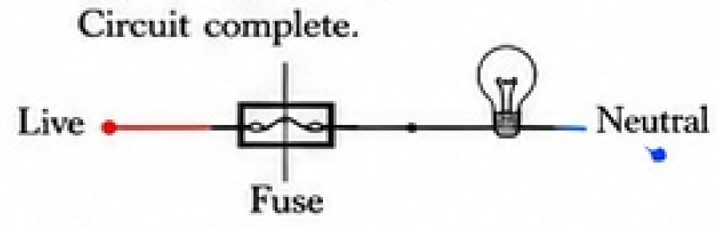
1. **Electric Fuse**
   - A fuse is a safety device used to protect circuits and electrical appliances from excessive current.
   - It contains a thin fuse wire that melts and breaks the circuit when current exceeds a safe limit.
2. **Working of Fuse**
   - When current is normal, fuse wire remains intact and circuit works.
   - When current exceeds the safe limit, heat produced melts the fuse wire and the circuit breaks, stopping the current.
3. **Why Fuse Wire Has Low Melting Point?**
   - Fuse wire is made of an alloy (tin–lead) that has low melting point and high resistivity.
   - Hence, it melts quickly on excessive current and protects the circuit.
4. **MCB (Miniature Circuit Breaker)**
   - MCB is an automatic electrical switch used to protect circuits from overcurrent and short circuit.
   - It can be reset (switched ON) after tripping.
5. **Working of MCB**
   - Under normal current, MCB remains ON.
   - If current exceeds the safe limit (due to overload or short circuit), MCB trips OFF automatically.
   - The mechanism uses a bimetal strip (overload) and electromagnetic coil (short circuit).
6. **Fuse vs MCB**
   - Comparison between reusable (MCB) and non-reusable (Fuse) devices.
7. **Short Circuit**
   - When live wire comes in direct contact with neutral wire (or earth) due to faulty insulation, a large current flows in a low-resistance path.
   - This is called short circuit.
8. **Overloading**
   - When too many high-power appliances are connected to one circuit, current exceeds the safe limit.
   - This is called **overloading.**
9. **Dangers of Short Circuit and Overloading**
   - May cause sparks, heating, damage to appliances.
   - Can melt wires and cause fire.
   - May lead to electric shock.
10. **Prevention and Safety**
    - Use proper protection devices (fuse/MCB).
    - Do not overload sockets.
    - Use wires of proper thickness and good insulation.
    - Regularly check wiring and appliances.
    - Follow household safety rules.

Diagram of an Electric Fuse

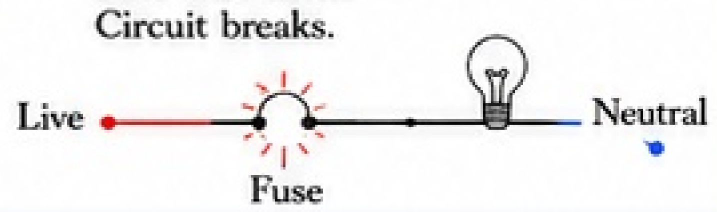


Working of Fuse

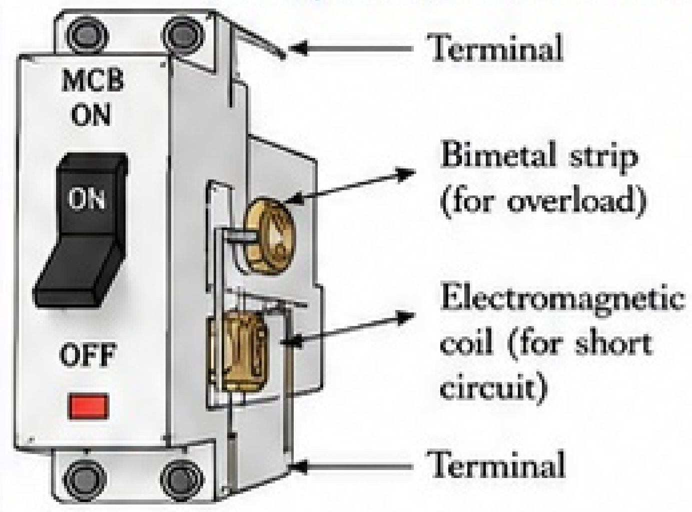
(i) Normal current:
Fuse wire intact → Circuit complete.



(ii) Excess current:
Fuse wire melts → Circuit breaks.



MCB (Miniature Circuit Breaker) – Concept



Working of MCB

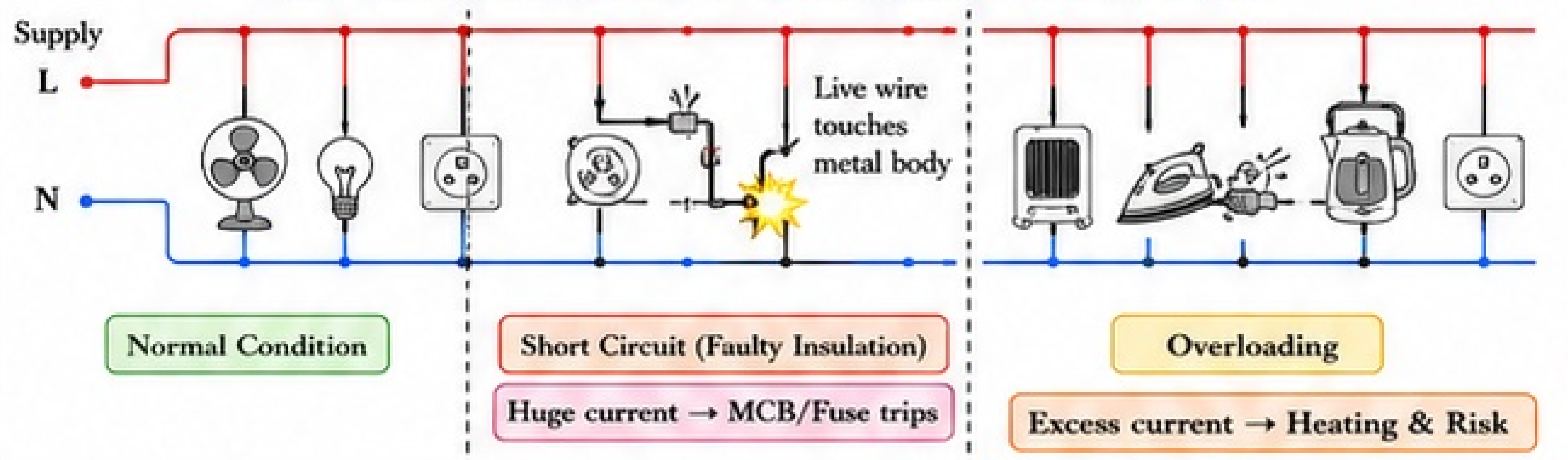
(i) **Normal current:** Bimetal is straight, coil is inactive. MCB remains ON.

(ii) **Overload:** Bimetal heats up and bends, releasing the latch → MCB trips OFF.

(iii) **Short circuit:** High current energizes the coil, attracts armature → MCB trips OFF instantly.

(iv) After removing the fault, MCB can be switched ON again.

Situation: Short Circuit and Overloading in a Household Circuit



Fuse vs MCB

Aspect	Fuse	MCB
Full Form	Electric Fuse	Miniature Circuit Breaker
Type	Non-automatic	Automatic
Operation	Melts and breaks circuit	Trips and breaks circuit
Usability	Not reusable	Reusable (resettable)
Response	Slower	Faster
Indication	No clear indication	ON/OFF position shows
Cost	Low	Higher
Maintenance	Needs replacement	No replacement needed
Use	Old installations	Modern installations

Causes and Prevention Table

Fault	Causes	Dangers	Prevention
Short Circuit	• Live and neutral wires touch each other • Damaged insulation • Water ingress • Loose connections	• Very high current • Sparks and heating • Fire hazard • Damage to appliances	• Use good insulation • Keep wires away from moisture • Proper installation • Use MCB/Fuse
Overloading	• Too many appliances in one socket • Using high-power appliances together • Old or undersized wiring	• Excess current • Heating of wires • Melting of insulation • Fire hazard	• Do not overload sockets • Use proper wiring & sockets • Use MCB/Fuse

QUICK RECALL

- Fuse melts on excess current.
- MCB trips automatically.
- Short circuit = live and neutral touch.
- Overloading = too many loads.
- Use protection devices for safety.

KEY IDEA

Protection devices break the circuit when current becomes unsafe. They protect life, property and appliances by preventing overheating and fire.

EXAM TIPS

- Draw labelled diagrams of fuse and MCB.
- Differentiate short circuit and overloading.
- Write causes, dangers and prevention in points.

CENTUM FOCUS

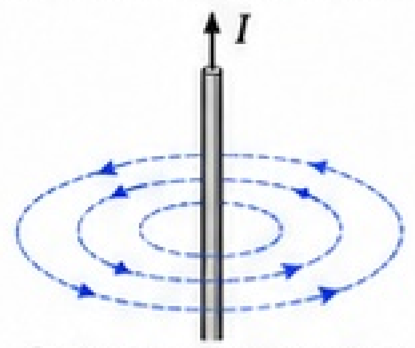
- Remember causes and prevention.
- Learn fuse vs MCB table thoroughly.
- Practice diagram-based questions.
- Relate to daily life situations.

Ch 12: Magnetic Effects of Electric Current

SECTION I - IMPORTANT DIAGRAMS, COMPARISONS, ANSWER FRAMES AND PRACTICE SET

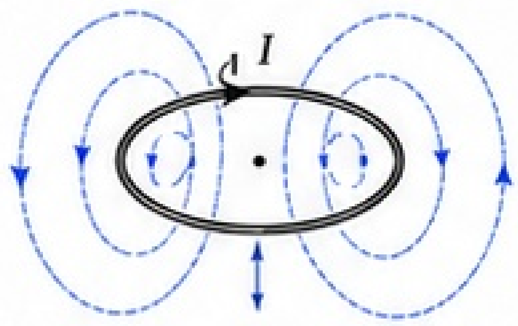
A. MUST-KNOW DIAGRAMS

1. Magnetic Field Around a Straight Current-Carrying Conductor



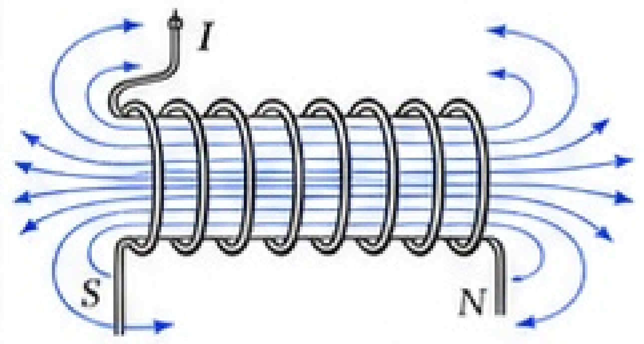
Field lines are concentric circles centered on the wire.

2. Magnetic Field Around a Circular Loop



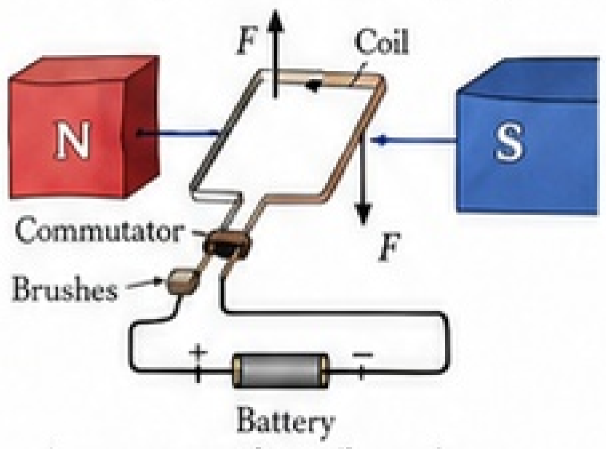
Field is strong and nearly uniform at the center of the loop.

3. Magnetic Field of a Solenoid



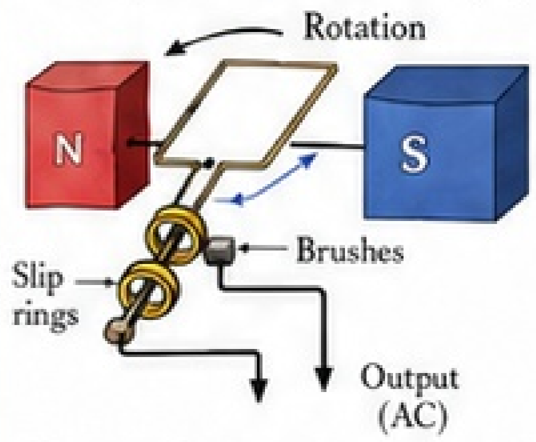
Inside the solenoid, field lines are nearly parallel and strong.

4. Simple Electric Motor (DC)



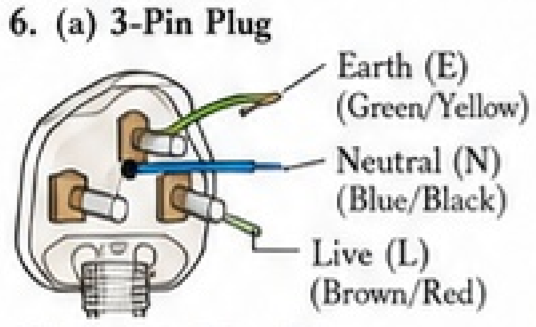
A current-carrying coil experiences a force and rotates continuously.

5. Simple Electric Generator (AC)



Rotation of the coil in a magnetic field induces an emf (AC output).

6. (a) 3-Pin Plug



(b) Fuse in Live Wire

Fuse protects the appliance by breaking the circuit in case of excess current.

B. FORMULA-FREE SUMMARY

- Electric current produces a magnetic field around a conductor.
- The direction of the field depends on the direction of current (Right-Hand Thumb Rule).
- Magnets have two poles: N and S. Like poles repel; unlike poles attract.
- Magnetic field lines are closed curves: outside N → S, inside S → N.
- A current-carrying conductor in a magnetic field experiences a force.
- **Motors:** Electrical energy → Mechanical energy (due to force on current).
- **Generators:** Mechanical energy → Electrical energy (due to electromagnetic induction).
- Domestic circuits use L, N, and E. Fuse in live wire ensures safety.

C. SHORT-ANSWER FRAME (BOARD STYLE)

1. State Right-Hand Thumb Rule.
- **Frame:** If you hold a current-carrying conductor in your right hand with the thumb pointing in the direction of current, the curled fingers show the direction of magnetic field lines.

2. State Fleming's Left-Hand Rule.
- **Frame:** Hold the thumb, forefinger and middle finger of your left hand mutually perpendicular.
Forefinger → Magnetic field (B),
Middle finger → Current (I).
Thumb → Motion/Force (F).

3. State Fleming's Right-Hand Rule.
- **Frame:** Hold the thumb, forefinger and middle finger of your right hand mutually perpendicular.
Forefinger → Magnetic field (B),
Thumb → Motion of conductor (M),
Middle finger → Induced current (I).

4. Explain working of an electric motor.
- **Frame:** A current-carrying coil placed in a magnetic field experiences forces on its two sides.
These forces form a couple which rotates the coil.
A split-ring commutator reverses the current every half turn, so rotation continues.

5. Explain working of an electric generator.
- **Frame:** When a coil is rotated in a magnetic field, the magnetic flux linked with it changes.
This induces an emf and current in the coil.
Slip rings and brushes collect the output (AC) from the rotating coil.

6. What safety precautions are used in domestic wiring?
- **Frame:** Use a 3-pin plug with proper earthing.
Connect fuse in the live wire.
Do not overload sockets. Use good insulation.
Never touch bare wires with wet hands.

D. COMPARISON TABLES

(i) Electric Motor vs Electric Generator

Aspect	Electric Motor	Electric Generator
Energy Conversion	Electrical → Mechanical	Mechanical → Electrical
Main Principle	Force on current-carrying conductor	Electromagnetic induction
Input	Electrical energy	Mechanical energy
Output	Mechanical energy	Electrical energy (AC)
Key Parts	Coil, commutator, brushes, magnet	Coil, slip rings, brushes, magnet
Direction of Current	DC (with commutator)	AC (from slip rings)
Example	Ceiling fan, mixer	AC generator in power plant

(ii) Fleming's Left-Hand Rule vs Fleming's Right-Hand Rule

Aspect	Left-Hand Rule	Right-Hand Rule
Use	Force on current-carrying conductor (Motor rule)	Induced current in moving conductor (Generator rule)
Thumb	Motion / Force (F)	Motion (M) of conductor
Forefinger	Magnetic field (B)	Magnetic field (B)
Middle finger	Current (I)	Induced current (I)
Application	Electric motor	Electric generator

E. PRACTICE SET (Answer in one line)

1. What is the shape of magnetic field lines around a straight conductor?
2. Which rule is used to find the direction of magnetic field around a current-carrying conductor?
3. Why are magnetic field lines called closed curves?
4. What happens when current in a conductor is increased?
5. Which parts of a simple DC motor help to keep it rotating?
6. Which principle is used in an electric generator?
7. Why is alternating current obtained in a generator?
8. What is the function of a fuse in a circuit?
9. Which wire is connected to the earth pin of a 3-pin plug?
10. Name one example each of electric motor and generator.

Answers (for self-check)

1. Concentric circles 2. Right-Hand Thumb Rule
3. They form closed loops: N → S outside, S → N inside.
4. Magnetic field becomes stronger.
5. Split-ring commutator and brushes.
6. Electromagnetic induction. 7. Due to changing direction of induced emf.
8. It melts and breaks the circuit in case of excess current.
9. Green/Yellow earth wire. 10. Motor: Mixer; Generator: Dynamo.

QUICK RECALL

- Current → Magnetic field.
- Right-Hand Thumb Rule for field direction.
- Motor: Force causes rotation.
- Generator: Rotation induces current.
- Fuse in live wire saves life.

KEY IDEA

- Electricity and magnetism are connected.
- Field, force and motion are interrelated.
- Rules help in remembering directions quickly.
- Safety in wiring is essential.

EXAM TIPS

- Draw neat, labeled diagrams.
- State rules clearly with finger directions.
- In comparisons, write in points.
- Use diagrams wherever possible in answers.

CENTUM FOCUS

- Revise rules and diagrams daily.
- Solve mixed practice questions.
- Differentiate motor, generator and rules clearly.
- Write crisp, point-wise answers in exams.

Ch 12: Magnetic Effects of Electric Current

SECTION J – MASTER REVISION SHEET, NCERT TRAPS AND CENTUM FOCUS

Key Definitions (At a Glance)

Term	Definition
Magnet	A material that produces a magnetic field and attracts magnetic materials.
Magnetic Field	The region around a magnet or current-carrying conductor where a magnetic force can be experienced.
Field Lines	Imaginary lines used to represent magnetic field; direction shows direction of field.
Solenoid	A long cylindrical coil of insulated wire. When current flows, it behaves like a bar magnet.
Electromagnet	A temporary magnet produced by a current-carrying coil, usually wound on a soft iron core.
Motor	A device that converts electrical energy into mechanical energy.
Generator	A device that converts mechanical energy into electrical energy.
Electromagnetic Induction	The phenomenon of inducing emf in a conductor by changing magnetic flux linked with it.
Fuse	A safety device made of low melting point wire; melts and breaks circuit when current is too high.
MCB	A circuit breaker that automatically trips off when current exceeds safe limit.
Earthing	Connecting the metal body of an appliance to the earth via a low resistance wire for safety.
Short Circuit	Accidental connection between live and neutral wires (or low resistance path), causing very high current.
Overloading	Using more electrical appliances than the circuit can safely handle, causing excessive current.

Master Summary – Whole Chapter

- **Electric current** produces magnetic effect around a conductor.
- Direction of magnetic field around a straight conductor is given by **Right-Hand Thumb Rule.**
- **Magnetic field lines** are concentric circles around a straight conductor; strength decreases with distance.
- Magnetic field inside a **solenoid** is strong and uniform like a bar magnet.
- **Electromagnets** are stronger than permanent magnets and their strength can be changed.
- **Electric motor** works on the principle that a current-carrying conductor placed in a magnetic field experiences a force, causing motion.
- Direction of force on conductor in motor is given by **Fleming's Left-Hand Rule.**
- **Generator** works on **electromagnetic induction**—changing magnetic flux induces emf and current.
- More turns, stronger magnetic field, faster rotation → more induced emf.
- Domestic electric circuits: **live wire, neutral wire** and **earthing wire.**
- Safety devices: **Fuse** and **MCB** protect from overcurrent.
- **Short circuit** and **overloading** are dangerous; may cause heating, fire and damage to appliances.
- **Proper earthing** prevents electric shock by providing a low resistance path for fault current.
- Understanding these concepts is essential for safe and efficient use of electricity.

NCERT Traps / Common Misconceptions

Misconception / Trap	Correct Explanation
Magnetic field lines start from North pole.	Field lines emerge from North pole and enter South pole outside the magnet; inside, they go from South to North.
Field lines intersect at a point.	Field lines never intersect; each point has only one direction of magnetic field.
No magnetic field exists inside a solenoid.	Inside a long solenoid, field is strong, uniform and parallel; outside, it is weak.
Strength of electromagnet depends only on current.	It depends on current AND number of turns AND nature of core material.
Motors work even if current direction is reversed.	Reversing either current or field reverses direction of force; both reversed → same direction.
Induced current needs battery.	Induced current is produced without battery by changing magnetic flux.
Fuse and MCB are same.	Fuse melts once and must be replaced; MCB resets after tripping.
Earthing is connected to neutral wire.	Earthing is connected to earth, not to neutral.
Short circuit is same as overloading.	Short circuit → very low resistance, very high current. Overloading → excessive load but not low resistance.
Magnetic field is stronger near poles.	For a bar magnet, yes; for a straight conductor, field is stronger near the conductor.

Concept Map – Whole Chapter

Magnetic Effects of Electric Current

- **Magnetic Field Due to Current**
  - Around Straight Conductor (R.H. Thumb Rule)
  - Solenoid and its Field
  - Magnetic Field Lines
  - Properties of Field Lines
- **Applications**
  - **Electric Motor**
    - Force on current-carrying conductor
    - Fleming's Left-Hand Rule
    - Converts electrical → mechanical energy
  - **Generator**
    - Electromagnetic Induction
    - Changing flux induces emf
    - Converts mechanical → electrical energy
  - **Electromagnet**
    - Coil + Soft Iron Core
    - Strength depends on current, turns, core and nature of core
    - Used in relays, cranes, bells
- **Electricity in Daily Life (Safety & Circuits)**
  - Domestic Circuits (Live, Neutral, Earth)
  - Safety Devices: Fuse, MCB
  - Hazards: Short Circuit, Overloading
  - Earthing: Safety from electric shock

Core Idea: Current → Magnetic Field → Force / Induction → Useful Devices and Safe Use of Electricity

10 Must Remember Points

1. **Right-Hand Thumb Rule** → Field around straight conductor.
2. **Field lines** are concentric circles around straight conductor.
3. **Inside solenoid** → strong and uniform field.
4. **Electromagnet** strength ∝ current, number of turns and nature of core.
5. **Fleming's Left-Hand Rule** → direction of force (motor).
6. **Generator** works on electromagnetic induction.
7. **More** turns, stronger field, faster rotation → more emf.
8. **Fuse** melts; **MCB** trips and can be reset.
9. **Earthing** protects from electric shock.
10. **Short circuit** and **overloading** are dangerous—avoid them.

Remember: Safety + Concepts + Practice = Centum in Science!

✓ Centum Checklist

- ☐ I can state and apply Right-Hand Thumb Rule.
- ☐ I can draw magnetic field lines for straight conductor and solenoid.
- ☐ I can explain the working of a motor.
- ☐ I can apply Fleming's Left-Hand Rule.
- ☐ I can explain the working of a generator.
- ☐ I can state factors affecting induced emf.
- ☐ I can define and compare electromagnet and permanent magnet.
- ☐ I can explain domestic electric circuit.
- ☐ I can differentiate fuse and MCB.
- ☐ I can explain earthing and its importance.
- ☐ I can explain short circuit and overloading.
- ☐ I can solve NCERT questions and Exemplar problems from this chapter.
- ☐ I can draw neat diagrams and label correctly.
- ☐ I can write key terms and definitions.

✓ Tick All – You are Centum Ready!

BEFORE EXAM – QUICK GLANCE

- Revise definitions and key rules.
- Redraw all important diagrams neatly.
- Solve NCERT Intext, Exercise and Exemplar questions.
- Write answers in points and use diagrams.
- Read NCERT "Do You Know?" and "Think and Discuss" boxes.
- Stay calm, confident and manage time smartly.

Aim High, Score Centum!